Calculus II — Integral Practice Problems

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Problems marked with ★ could be done more easily with partial fractions, but you should still attempt these problems as there could be algebraic detours. I will try my best to give solutions that require some cleverness but not partial fraction.

Tier 1 — Warm Up (2 problems)

1. $\displaystyle\int \frac{3x+1}{x^2+2x+5}\,dx$

2. $\displaystyle\int \frac{1}{x^2\sqrt{x^2-2}}\,dx$

Tier 2 — Tricky (8 problems)

3. $\displaystyle\int \frac{x^2}{(x-1)(x^2+1)}\,dx$ ★

4. $\displaystyle\int \frac{1}{1+\sin x}\,dx$

5. $\displaystyle\int \ln(1+x^2)\,dx$

6. $\displaystyle\int \frac{\sqrt{x}}{1+x}\,dx$

7. $\displaystyle\int \sec^3 x \tan^4 x\,dx$

8. $\displaystyle\int \frac{x+1}{x^2+x+1}\,dx$

9. $\displaystyle\int \frac{e^{2x}}{e^x+1}\,dx$

10. $\displaystyle\int x \arctan x\,dx$

Tier 3 — Challenging (8 problems)

11. $\displaystyle\int \frac{1}{x^4+1}\,dx$ ★

12. $\displaystyle\int \frac{1}{\sqrt{e^x+1}}\,dx$

13. $\displaystyle\int_0^{\pi/2} \frac{\sin x}{\sin x+\cos x}\,dx$

14. $\displaystyle\int \frac{x\arctan x}{(1+x^2)^2}\,dx$

15. $\displaystyle\int_0^1 \frac{\ln(1+x)}{1+x^2}\,dx$

16. $\displaystyle\int \frac{1}{\sqrt{x}+\sqrt[3]{x}}\,dx$

17. $\displaystyle\int \frac{\sin x}{\sin x+\cos x+1}\,dx$ ★

18. Show that $\displaystyle\int_0^1 x^m(1-x)^n\,dx = \frac{m!n!}{(m+n+1)!}$, for $n, m = 0, 1, 2, \cdots$. This is known as the Beta function.

Problems 19–39 (Extra) Note that the difficulty level for this section is unordered.

Algebraic / Partial Fraction Challenges

19. $\displaystyle\int \frac{1}{1+x^3}\,dx$ ★

20. $\displaystyle\int \frac{x^4+1}{x^6+1}\,dx$ ★

21. $\displaystyle\int \frac{x^2+1}{x^4+x^2+1}\,dx$ ★

22. $\displaystyle\int \frac{1}{x(x^4+1)}\,dx$ ★

Substitution Ingenuity

23. $\displaystyle\int e^{\sqrt{x}}\,dx$

24. $\displaystyle\int \sqrt{\tan x}\,dx$ ★

25. $\displaystyle\int \frac{x^2-1}{x^4+3x^2+1}\,dx$

Trig Integral Tricks

26. $\displaystyle\int \frac{1}{\sin^4 x+\cos^4 x}\,dx$

27. $\displaystyle\int \frac{\sin^2 x}{1+\sin x\cos x}\,dx$

28. $\displaystyle\int \frac{\cos x}{\sin^2 x+\sin x+1}\,dx$

29. $\displaystyle\int \frac{1}{(\sin x+\cos x)^4}\,dx$

Definite Integral Symmetry

30. $\displaystyle\int_0^{\pi/4} \ln(1+\tan x)\,dx$

31. $\displaystyle\int_0^{\pi} \frac{x\sin x}{1+\cos^2 x}\,dx$

32. $\displaystyle\int_0^{\pi/2} \frac{1}{1+\tan^n x}\,dx \quad (n\in\mathbb{Z}^+)$

33. $\displaystyle\int_0^{\pi/2} \ln(\sin x)\,dx$

34. $\displaystyle\int_0^{\pi/2} \frac{\sin^2 x}{\sin x + \cos x}\,dx$

Integration by Parts / Reduction

35. $\displaystyle\int_0^1 x^3(\ln x)^2\,dx$

Combining Multiple Techniques

36. $\displaystyle\int_0^\infty \frac{x^2}{(x^2+1)^2}\,dx$

37. $\displaystyle\int \frac{1}{(x^2+1)\sqrt{x^2+4}}\,dx$

38. $\displaystyle\int \frac{x^2 \arctan x}{1+x^2}\,dx$

39. $\displaystyle\int \sqrt{2\sqrt{x} - x}\;dx$